

Let i, j, n be positive integers such that $0 < i < j < n$. Define

$$p^* = \frac{\binom{n}{i}^{\frac{1}{j-i}}}{\binom{n}{i}^{\frac{1}{j-i}} + \binom{n}{j}^{\frac{1}{j-i}}}.$$

I have three questions. Here are the first two:

- (1) Is $i < np^* < j$?
- (2) Is $2np^* < i + j$?

If it helps, you may impose the restriction on i, j, n that implies $p^* > 1/2$.

Next consider the binomial probability distribution consisting of the probabilities of k successes in n trials, where the probability of a success is denoted by p , ranging from 0 to 1. Then the binomial probabilities are given by

$$f_p(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Now rearrange these probabilities into increasing order and define r_k to be the rank of $f_p(k)$ among these $k + 1$ rearranged probabilities. Thus if there are no ties among the binomial probabilities, $r_0, r_1, \dots, r_n$ is a permutation of $1, 2, \dots, n + 1$.

Now exclude the cases $p = 0, 1/2, 1$, so there can be at most one tie among the binomial probabilities, since these probabilities are unimodal. If there is a tie, say $f_p(i) = f_p(j)$ for $i < j$, then define $r_j = r_i + 1$, so again $r_0, r_1, \dots, r_n$ is a permutation of $1, 2, \dots, n + 1$.

Finally, define $w_k = r_k - 1$ for $k = 0, 1, \dots, n$, so $w_0, w_1, \dots, w_n$ is a permutation of $0, 1, \dots, n$.

Now consider my third question (3) appearing below: Suppose p is such that there is a tie at i and j , that is, $f_p(i) = f_p(j)$ for $i < j$. Then it is easy to show that

$$p = p^* = \frac{\binom{n}{i}^{\frac{1}{j-i}}}{\binom{n}{i}^{\frac{1}{j-i}} + \binom{n}{j}^{\frac{1}{j-i}}}.$$

- (3) Is $w_i(i + j - 2np^*) + (j - np^*) > 0$?